

Natural Language Processing

KogSys-KI-B: Introduction to Artificial Intelligence

...based on Prof. Stefan Ultes' slides, with his kind permission

Outline



1. What is Natural Language Processing?
2. Historic Approaches
   2.1 Georgetown IBM Experiment
   2.2 Eliza
3. Information Retrieval: Vector Representations
4. Text Prediction and Language Generation
5. Speech Recognition
6. Natural Language Understanding
7. Text Classification

What is Natural Language Processing?



- Sub-field of Artificial Intelligence
  - How can a computer deal with natural (i.e. human) language
  - How to extract key information from text and how to process it

- Manning and Schütze (1999):

> "[...] understanding the relationship between utterances and the world,
> and [...]
> understanding the linguistic structures by which language communicates"

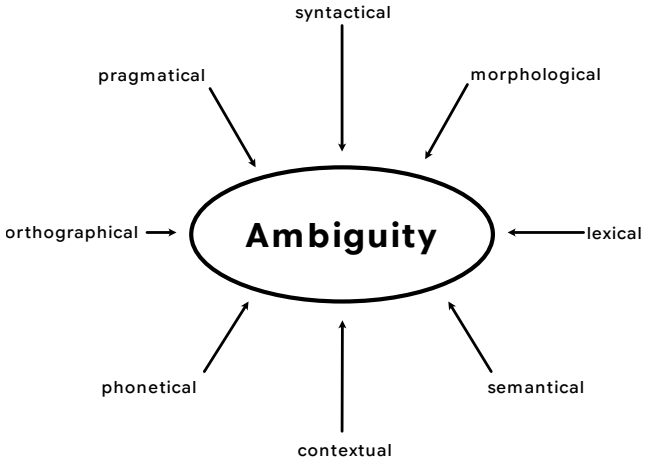
What is Natural Language Processing?



- Alan Turing
  - Language as indicator of thinking ability
  - Turing test operational test for intelligence

- NLP: Methods to resolve ambiguity:
  Meaning of “I made him duck.” (Jurafsky & Martin, 2026)
  1. I cooked waterfowl for him.
  2. I cooked waterfowl belonging to him.
  3. I created the (plaster) duck he owns.
  4. I caused him to quickly lower his head or body.
  5. I waved my magic wand and turned him into undifferentiated waterfowl.

Challenge in NLP: Ambiguity



What is Natural Language Processing?



- Require knowledge: knowledge of language
  - phonetical knowledge
  - lexical knowledge (what is a word)
  - syntax knowledge (word order)
  - semantic and pragmatic knowledge (what is the meaning)
  - discourse knowledge

Georgetown IBM Experiment

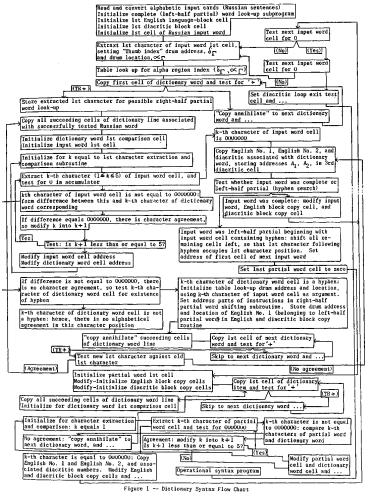


Figure 1 — Dictionary Syntax Flow Chart

- First NLP project
- “Mechanical Translation”: Russian to English
- Rule-based approach taking into account:
  - Word meaning
  - Word order

Ornstein, J. (1955). Mechanical Translation: New Challenge to Communication. *Science*, *122*(3173), 745–748. https://doi.org/10.1126/science.122.3173.745

Georgetown IBM Experiment



SENTENCE: **vyelyichyina ugla opryedyelyayetsya otnoshyenyiyem dlyini dugi k radyiusu.**

ANALYSIS:

RUSSIAN WORD	ENGLISH EQUIVALENTS		1st	2nd	3rd	RULE
	I	II	CODE	CODE	CODE	NO
vyelyichyina	magnitude	---	***	***	**	6
ugl-	coal	angle	121	***	25	2
-a	of	---	131	222	25	3
opryedyelyayestsya	is determined	---	***	***	**	6
otnoshyenyi-	relation	the relation	151	***	**	5
-yem	by	---	131	***	**	3
dlyin-	length	---	***	***	**	6
-i	of	---	131	***	25	3
dug-	arc	---	***	***	**	6
-i	of	---	131	***	25	3
k	to	for	121	***	23	2
radyius-	radius	---	***	221	**	6
-u	to	---	131	***	**	3

TARGET
SENTENCE: **magnitude of angle is determined by the relation of length of arc to radius.**

Algorithm:

1. Translate Russian words into numerical codes
2. Perform case-analysis on each numerical code
   - Choose between possible English word translations
   - Reorder English words
   - Omit English words

Ornstein, J. (1955). Mechanical Translation: New Challenge to Communication. *Science*, *122*(3173), 745–748. https://doi.org/10.1126/science.122.3173.745

Eliza (1966)

- Text-based simulation of a Rogerian psychotherapist
- Considered first Chatbot
- Simple program: uses rules and templates (designed by experts)
- Input processing: decomposition rules triggered by keywords
- Responses: reassembly rules associated with decomposition rules
- Demo: https://www.masswerk.at/eliza/

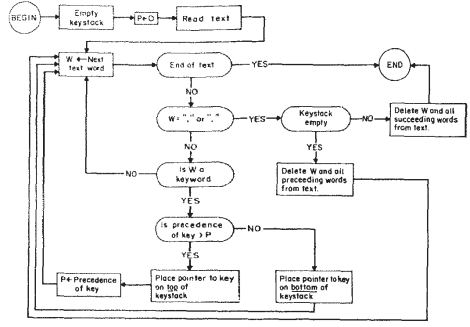


FIG. 2. Basic flow diagram of keyword detection

Weizenbaum, J. (1966). ELIZA—a computer program for the study of natural language communication between man and machine. *Communications of the ACM, 9*(1), 36–45. https://doi.org/10.1145/365153.365168

Eliza (1966): Code Example



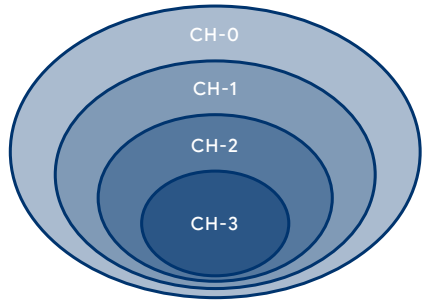
```
["dream", 3, [
    ["*", [
        "What does that dream suggest to you ?",
        "Do you dream often ?",
        "What persons appear in your dreams ?",
        "Do you believe that dreams have something to do with your problem ?"
    ]]
]],
["perhaps", 0, [
    ["*", [
        "You don't seem quite certain.",
        "Why the uncertain tone ?",
        "Can't you be more positive ?",
        "You aren't sure ?",
        "Don't you know ?",
        "How likely, would you estimate ?"
    ]]
]],
```

Rule-based Approaches: Pattern Matching



- Select response/behavior based on if-then-else rules
- Condition: match on linguistic patterns
- Tools:
  - Keywords
  - Regular Expressions (CH-3)[1]
  - Context-free grammars (CH-2)[1]
  - Context-sensitive grammars (CH-1)[1]

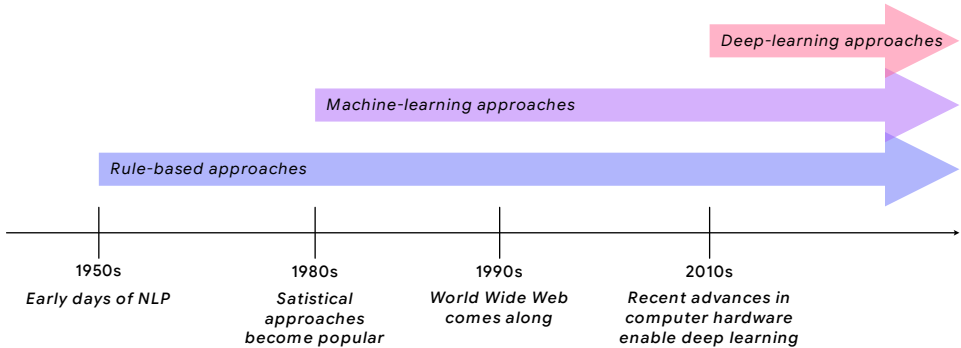
→ Parsing



[1] Chomsky Hierarchy

Timeline of NLP Approaches

- Rule-based: require no data, leverage expert knowledge, guarantees about performance
- Statistical: require data, applicable where complex rules are hard to find, sometimes wrong



Kochmar, E. (2022). *Getting Started with Natural Language Processing.* Simon and Schuster



What are NLP applications you use in your everyday life?

Example NLP Applications

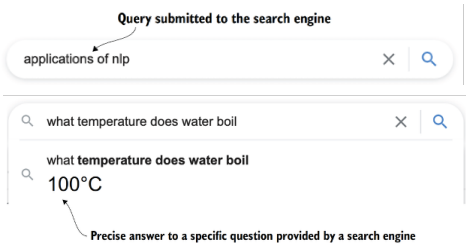


- Information Search/Question Answering
- Text prediction and language generation
- Conversational Agents and intelligent virtual assistants
- Spam filtering
- Machine translation
- Spell- and grammar checking

Information Retrieval



- Goal: find documents/websites/specific information given a search query
- Examples
  - All documents on your computer mentioning “management meetings”
  - All websites about “applications of nlp”
  - Answer to a particular question



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Information Retrieval



- Processing steps:
  1. Submit “query”—the search question, phrase or words
  2. Search engines provides concrete answers (in case of specific questions) or set of results
  3. Results are sorted in the order of relevance and returned
- Key question: how to define/derive relevance?

Information Retrieval

Finding documents: binary decision



If a document contains “meeting” and “management” keep it, otherwise discard it

Information Retrieval



Scoring relevance

- Humans:
  - Read through documents
  - Check with internal “representation”, i.e. contextual concept of search term

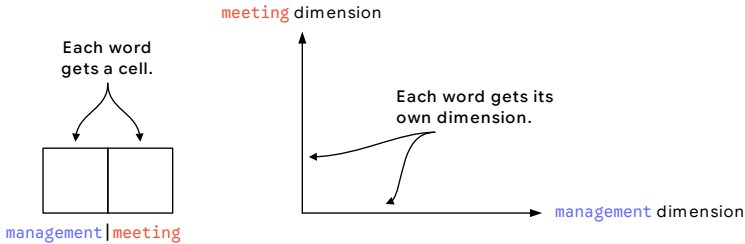
  $\rightarrow$ machines do not have this internal representation
- Machines:
  - Use *numerical representation*, i.e. vectors, arrays, ...
  - Convert query and documents to numerical representation
  - Use *distance* to score relevance

Vector Representations: Bag of Words



Idea: Represent documents/queries as a numerical vector by counting word occurrences

Query: `"management meeting"`



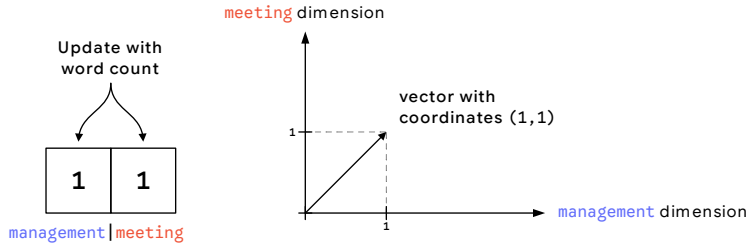
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Vector Representations: Bag of Words



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Query: "management meeting"



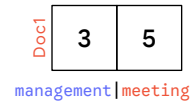
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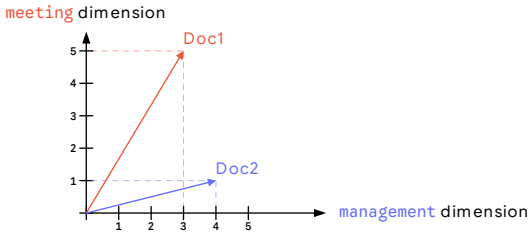
Vector Representations: Bag of Words



Idea: Represent documents/queries as a numerical vector by counting word occurrences

```
Doc1 = "meeting ... management ... meeting ... management ... meeting ...
    management ... meeting ... meeting"
Doc2 = "management ... meeting ... management ... managemnt ... management"
```





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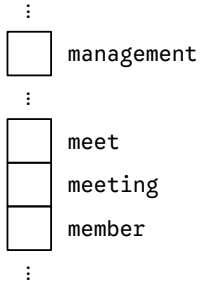
Vector Representations: Bag of Words



Idea: Represent documents/queries as a numerical vector by counting word occurrences

In practice:

- High-dimensional
- Pre-defined vocabulary, each word one dimensional
- Representations of *single words* are called 1-hot encodings: the dimensions of the word carries a 1, the rest 0



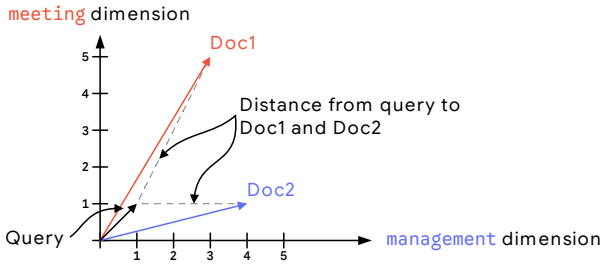
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Information Retrieval



Idea: Use similarity/distance in shared vector space to derive relevance

- Euclidean distance
- Cosine similarity

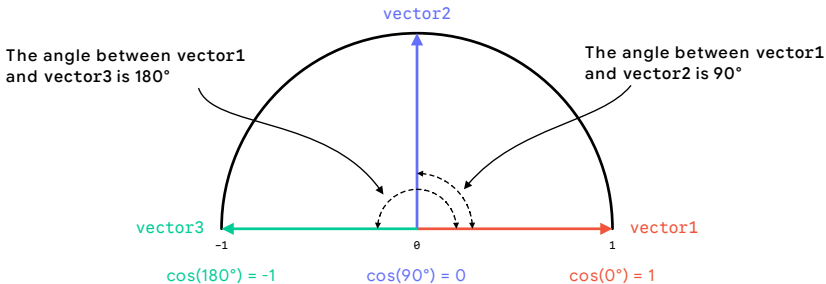


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Cosine Similarity

Idea: Define similarity by the angle between vectors

→ Similarity not affected by vector length



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Vector Representations: Word Embeddings



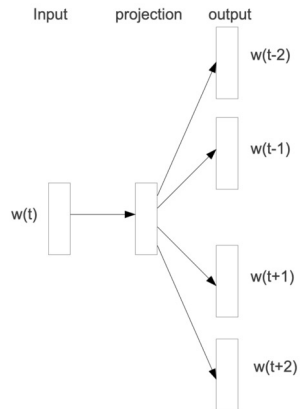
- Idea: Represent words as a numerical vector by learning vector representations
- Interesting Observation: Words with similar meaning appear in the same context of words
- Approach:
  - Map words to vectors in $\mathbb{R}^N$ for large $N \in (500, 2000)$
  - Apply optimization techniques to find a mapping from words to vectors that $d(w_1, w_2) \sim d(c_1, c_2)$ where c_i is the lexical context of w_i

  $\rightarrow$ Semantically similar words appear in similar contexts, thus if the contexts are close to each other, the words are also close to each other
- Research in NLP explores different realizations of word embeddings (word2vec, BERT, GPT, ...) and utilizes them for different tasks

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Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed Representations of Words and Phrases and their Compositionality. *Advances in Neural Information Processing Systems, 26*

Vector Representations: Word Embeddings



Word2vec (Mikolov et al., 2013):

- Word representation in semantic space
- Word vectors learned from data
- Skip-gram: predicting surrounding words
- Training loss:

$$\frac{1}{T}\sum_{t=1}^{T}\sum_{-c\leq j\leq c,\ j\neq 0}\log\ p(w_{t+j}|w_t)$$

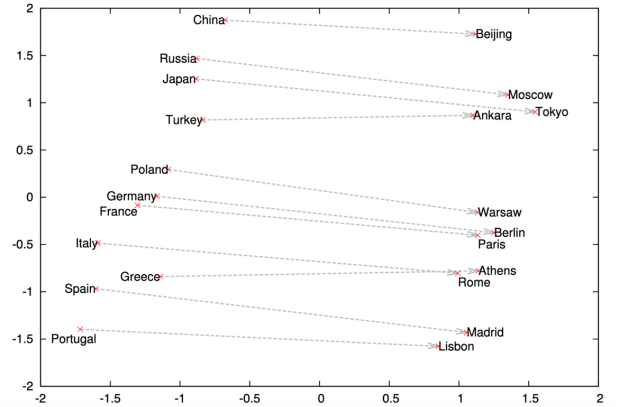
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Vector Representations: word2vec



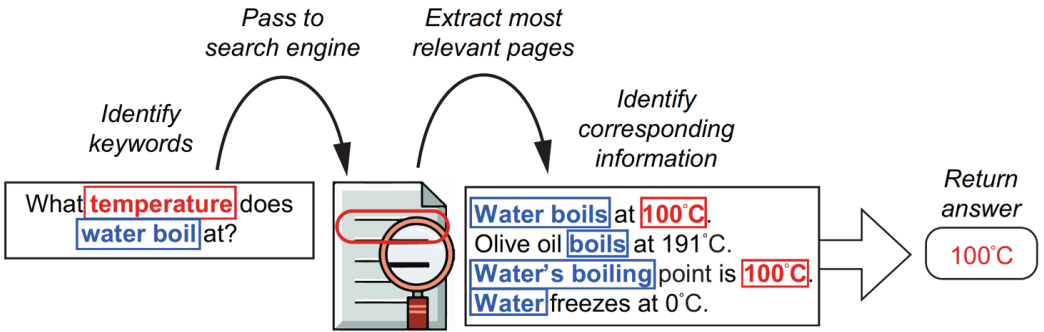
Implicit learning of word relationships

$$? = \mathrm{vec}(\textit{“Berlin”}) - \mathrm{vec}(\textit{“Germany”}) + \mathrm{vec}(\textit{“France”})$$



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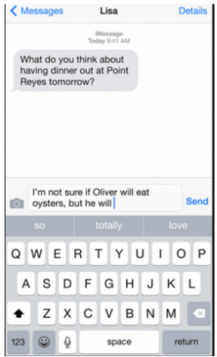
From Information Retrieval to Question Answering



Text Prediction and Language Generation



- Smartphone: predictive keyboard
- Auto email feature





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Text Prediction and Language Generation



Humans predict language, too!

- Complete the word: “langu”
- Continue the partial sentence “natural language”
- “For exmaple, you might miss some spleling miskates in this sentence.” (Kochmar, 2022)

→ make use of probabilities of character and word sequences

Text Prediction and Language Generation



- Probability of a word sequence:

$$P(w_1, w_2, \ldots, w_t)$$

- Total probability calculated from conditional probability of next word:

$$P(w_t | w_1, w_2, \ldots, w_{t-1})$$

- This is called *(Statistical/Probabilistic) Language Model*

Text Prediction and Language Generation



Count and divide?

$$P(\textit{nice}|\textit{the weather is}) = \frac{\text{count}(\textit{the weather is nice})}{\text{count}(\textit{the weather is})}, \qquad P(X|Y) = \frac{P(X,Y)}{P(Y)}$$

- No, too many possibilities!
- Context is important, but may introduce too many constraints

→ Approximation:

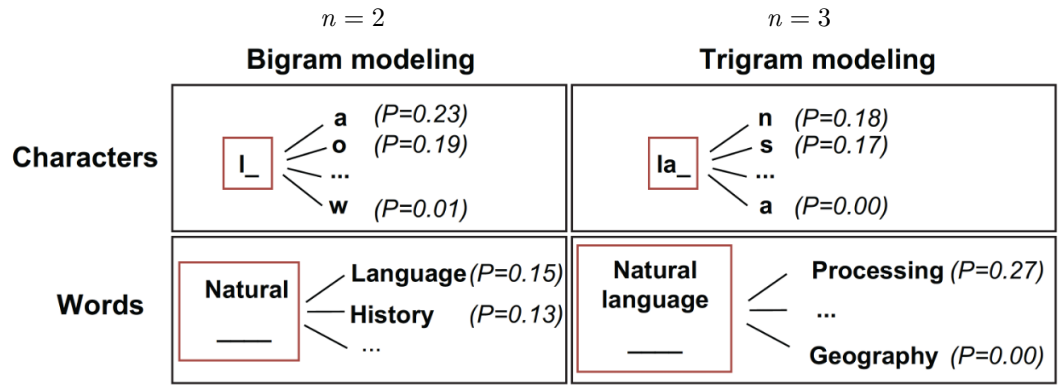
$$P(w_t|w_1, w_2, w_3, \ldots, w_{t-1}) \approx P(w_t|\underbrace{w_{t-n+1}, \ldots, w_{t-1}}_{\text{count: } n})$$

→ n-gram

Markov Assumption[1]

[1] The Strange Math That Predicts (Almost) Anything, Veritasium.

Text Prediction and Language Generation



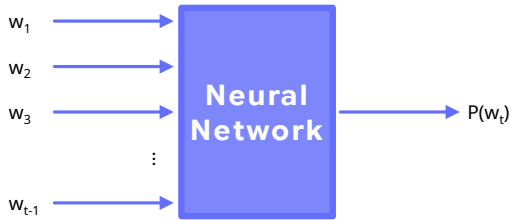
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Text Prediction and Language Generation



- Neural Language Models

$$P(w_t|w_1, w_2, \ldots, w_{t-1})$$



- Neural Network architectures:
  - Recurrent Networks
  - Transformer Networks

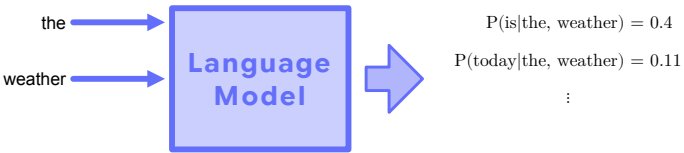
Examples for neural language models:
GPT-5, Gemini, Claude, etc.

Text Prediction and Language Generation



Language generation:

- Use probabilities to select next word (e.g. $\max$, sampling)
- Add generated word iteratively to condition

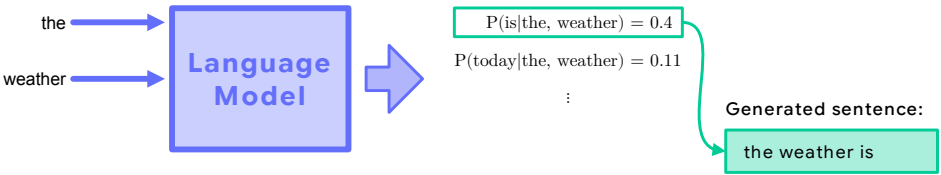


Text Prediction and Language Generation



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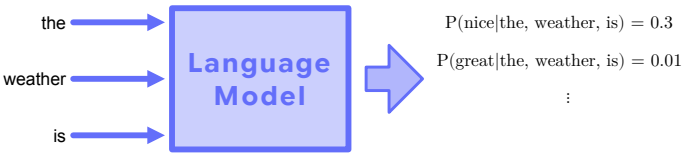


Text Prediction and Language Generation



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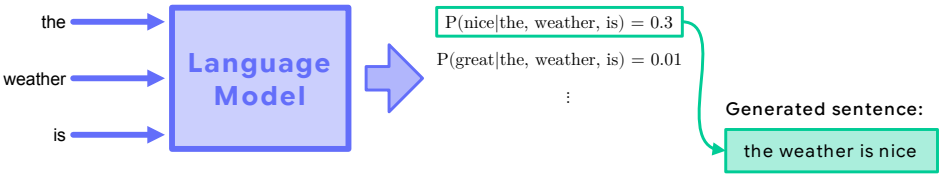


Text Prediction and Language Generation



Language generation:

- Use probabilities to select next word (e.g. $\max$, sampling)
- Add generated word iteratively to condition



Text Prediction and Language Generation



Some other use of language models

- Machine Translation

$$P(\textbf{high} \text{ winds tonight}) > P(\textbf{large} \text{ winds tonight})$$

- Spell correction
  The office is about fifteen **minuets** from my house

$$P(\text{about fifteen } \textbf{minutes} \text{ from}) > P(\text{about fifteen } \textbf{minuets} \text{ from})$$

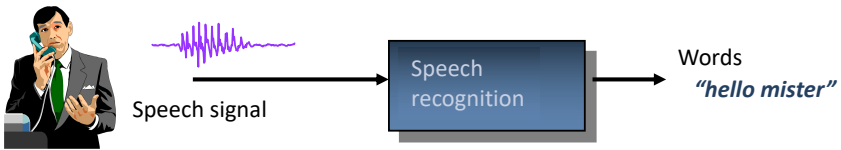
- Speech Recognition

$$P(\text{I saw a van}) \gg P(\text{eyes awe of an})$$

Speech Recognition

What is Speech Recognition?

Speech recognition automatically extracts the string of spoken words from the speech (audio) signal

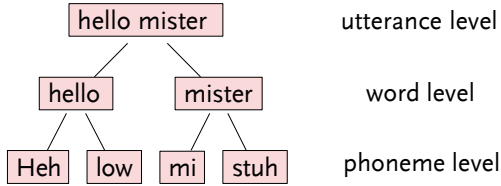


Speech Recognition



Hierarchical approach combining bottom-up (recognition) and top-down (generative) approaches

- starting with an acoustic level: phonemes
- syllable level, word level
- utterance (sentence) level



Speech Recognition



Speech Recognition Problem

- Find the most likely sequence W of words given a sequence X of acoustic feature vectors

$$\arg\max_W P(W|X)$$

- Bayes' law to create generative model

$$\arg\max_W P(W|X) = \arg\max_W \frac{P(X|W)P(W)}{P(X)} \approx \arg\max_W P(X|W)P(W)$$

$\rightarrow$ Acoustic model $P(X|W)$ and language model $P(W)$.

Speech Recognition



Automatic Speech Recognition (ASR)

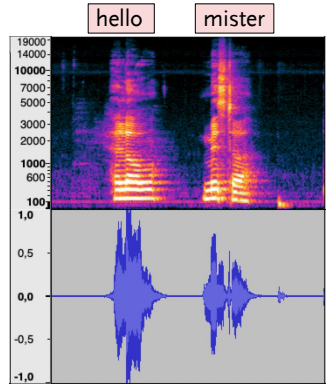
- Acoustic model:
  $\rightarrow$ what is the probability to observe acoustic features X if the word sequence is W
- Linguistic/Language model:
  $\rightarrow$ what is the prior probability that the word sequence W occurs (e.g. n-grams)
- Search/Decoding:
  $\rightarrow$ find the word sequence with the highest probability

Speech Recognition



Acoustic Modeling Based on Phonemes

- *Phonemes* are minimal classes of sounds that serve distinct functions in speech acts
  - phonemes can best be analyzed in the *frequency domain*, i.e., sound as a function $time \mapsto frequency$, not $time \mapsto sound\ pressure\ level$ as recorded in raw audio data
  - *pattern recognition* for identifying phonemes
- Problems arising from variations in pronunciation
  - some phonemes may not be uttered/recognized
  - mix-up of phonemes occurs

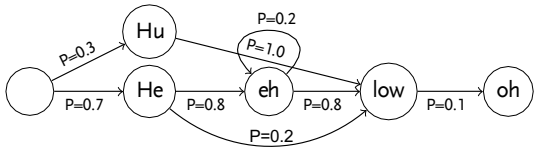


Speech Recognition



Hidden Markov Models

- Words are modeled as sequences of phonemes (similar to finite automata)
- Speech recognition as sequence recognition task in a probabilistic model: *Hidden Markov Model (HMM)* (details e.g. in KogSys-ML-B)



- The *Viterbi Algorithm* allows to compute the probability that a word X^* (i.e. phoneme sequence) represented as an HMM has been uttered, given a probabilistic model of phoneme utterance λ and their recognition $\mathcal{O}$.

Natural Language Understanding



- Goal: interpret a word sequence to derive meaning as structured data
- Meaning representation: semantic frames are a simple formalism to encode meaning
  - Intent: a semantic concept that describes “what the user wants” from a system with respect to the system task
  - Slots and values: parameters to further specify the intent
  - Domain: topical grouping of intents

Natural Language Understanding



Semantic Frame Example

I want to take a train from Bamberg to Nuremberg on December second.



```
train-information:book_train(data=02-12-26, place-from=Bamberg, place-to=Nuremberg)
```

domain — intent — semantic slots and values







Text Classification — Information Extraction

Text Classification



- Goal: assign a label to text (document, smaller text segments, utterances, ...)
- Examples:
  - Domain classification
  - Intent recognition
  - Spam filtering
  - Sentiment analysis
  - Topic classification
  - ...

Text Classification



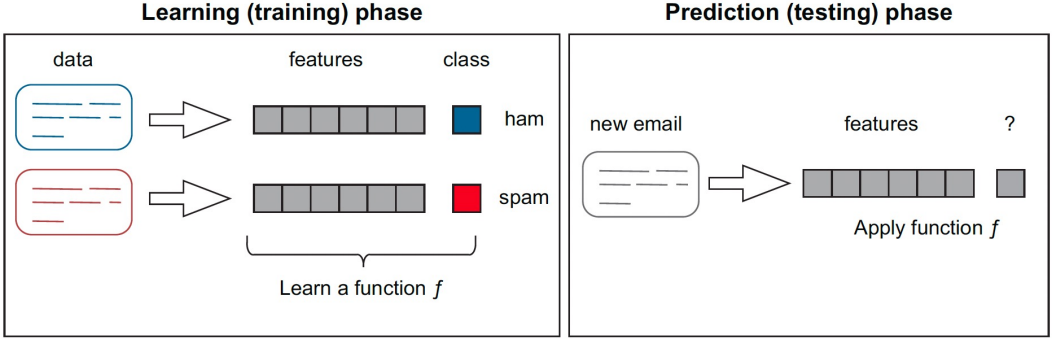
1. Text is represented through feature vectors
   - Bag-of-words
   - Embedding-based representations
2. A statistical model is trained on examples for each class, e.g. *spam* vs. *ham*[1], to learn a function f from text to the respective class, i.e. supervised learning
3. During application, the model is used as the decision function for new text to be classified

In other words: Turn text into a representation suitable as input to some machine learning model

[1] This is the common term used for non-spam

Text Classification

Example: Spam Filtering



Summary



- Natural Language Processing (NLP) uses AI approaches to process natural language
- Omni-present challenge: ambiguity
- There is a plethora of different tasks that make use of NLP, many of them are already used in our daily lives, most of them we don't notice or think about
- Pattern Matching can be a suitable approach for simple problems
- Advanced methods rely on vector representations of words and text as input to machine learning algorithms
- Word embeddings are concise representations of a word based on its context
- Language Models are an important foundational method and used for many NLP tasks

NLP an der Uni Bamberg



Generation Dialogue

Lehrstuhl für Sprachgenerierung und Dialogsysteme (Prof. Stefan Ultes)

- DS-IDS-B: Einführung in Dialogsysteme
- DS-ConvAI-M: Andvanced Dialogue Systems and Conversational AI

BamNLP

Lehrstuhl für Grundlagen der Sprachverarbeitung (Prof. Roman Klinger)

- NLProc-ALV-B: Algorithmisches Sprachverstehen
- NLProc-IRTM-B: Information Retrieval and Text Mining
- ...diverse Masterveranstaltungen

...und selbstverständlich Seminare und Projekte an beiden Lehrstühlen